

Jacob Cyrus Booeshaghi

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EDUCATION

University of Florida, Honors Program

Bachelor of Science in Biomedical Engineering, expected graduation Spring 2027

GPA: 3.92, Dean's List: Fall 2023, Spring 2024, Fall 2024, Spring 2025

Relevant Coursework: Computer Aided Graphics and Design (SolidWorks), Medical 3D Printing, Biomaterials

EXPERIENCE

Florida Advanced Manufacturing and Systems Integration Laboratory

Undergraduate Researcher, January 2025 - Present, ~10 hours a week

- Designed and executed 3D bioprinting experiments to fabricate perfusable tissue constructs using sacrificial channeling within hydrogel-based bioinks; validated channel integrity through dye perfusion and quantitative image analysis (ImageJ)
- Supporting ongoing development of a microfluidic lab-on-a-chip (LOC) system for Type 1 diabetes research through CAD design, SLA prototyping, and literature review on microfluidics and sensing approaches
- Prepared and tested various shear-thinning gelatin microgel-based bioinks in 3D bioprinting experiments.

GRiP (Generational Relief in Prosthetics)

Design Team Member, Spring 2024 - Spring 2025

- Collaborated on the design and fabrication of a pediatric prosthetic arm, contributing to mechanical design, prototyping, and assembly
- Manufactured and refined 3D-printed components to improve fit, durability, and functional performance
- Assembled and tested prosthetic systems, verifying mechanical functionality and component integration
- Integrated EMG sensors with servo actuation systems to enable user-controlled prosthetic movement

SKILLS

- Programming: Python, MATLAB, Java, Arduino
- CAD & Design: SolidWorks (CSWA Certified), Mimics and 3-Matic
- Prototyping & Fabrication: 3D printing, machining (mill, lathe), welding, casting, blacksmithing
- Testing & Analysis: ImageJ, experimental design, data analysis

LEADERSHIP

Gator Blacksmithing

President, May 2025 - Present

- Lead 50+ members in a hands-on fabrication environment, overseeing forge operations, safety protocols, and technical skill development
- Designed and implemented a structured, mastery-based training system focused on fabrication techniques, process control, and safety
- Organized and managed workshops and forging sessions, coordinating resources, equipment, and compliance with university safety standards
- Led design and fabrication of a Damascus steel Bowie knife for the TMS 2026 Bladesmithing Competition

Vice President, January 2024 - May 2025

- Coordinated fabrication efforts for the FIERF 2025 national competition, contributing to the design and production of an award-winning forged hammer (\$2000 prize)

UF ASME Biodesign Exoskeleton Team

Control Systems Subteam Lead, Spring 2025 - Present

- Lead development of a real-time control system for a joint-actuated exoskeleton using OpenSim gait data
- Designed algorithms to track joint kinematics, compare real-time motion to target gait cycles, and adjust actuator output for assistive torque generation
- Contribute to system-level integration of sensing, control, and actuation for assistive mobility devices